Simple Even-Case Constructions in Knuth's Cycles and Their Gemini-Generated Proofs*

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Abstract

We study the Hamiltonian decomposition problem for the directed Cayley graph $\Gamma_m = \mathrm{Cay}(\mathbb{Z}_m^3, \{e_1, e_2, e_3\})$, which is a problem that came up when Knuth was writing a future volume of *The Art of Computer Programming*. The odd m case admits a simple construction together with a complete proof [6, 1], so the remaining difficulty lies in the even m case. A recent work [1] identified a simple even-case construction and computationally verified it for all even $m \leq 2000$, but left open a symbolic proof of correctness for all m. In this note, we present an alternative simple construction, and accompanying files contain complete AI-generated proof drafts both for this construction and for the construction of [1].

1 Introduction

We study the Hamiltonian decomposition problem for the directed Cayley graph

$$\Gamma_m = \mathrm{Cay}(\mathbb{Z}_m^3, \{e_1, e_2, e_3\}),$$

where e_1, e_2, e_3 are the three standard basis directions modulo m. Thus the vertex set is $\mathbb{Z}_m^3$, and from each vertex (i, j, k) there are exactly three outgoing arcs,

$$(i, j, k) \to (i + 1, j, k), \qquad (i, j, k) \to (i, j + 1, k), \qquad (i, j, k) \to (i, j, k + 1) \pmod{m}.$$

Since Γ_m has m^3 vertices and out-degree 3, it has $3m^3$ directed arcs in total. The Hamiltonian decomposition problem asks for which integers $m > 2$ these arcs can be partitioned into three directed Hamiltonian cycles, each of length m^3. Equivalently, one may ask whether the three outgoing arcs at every vertex can be assigned to three color classes so that each color class forms a directed Hamiltonian cycle spanning all vertices of Γ_m.

This problem arose while Knuth was writing about directed Hamiltonian cycles for a future volume of *The Art of Computer Programming* [6]. He had solved the case $m = 3$ and posed the general Hamiltonian decomposition problem as an exercise; after Filip Stappers empirically found solutions for $4 \leq m \leq 16$, he posed the problem to Claude Opus 4.6, which, after 31 explorations with substantial human coaching, eventually found a construction for all odd m. Knuth then verified the construction, proved it correct, and published the result, identifying exactly 760 valid "Claude-like" decompositions for odd $m > 1$; the even case, however, remained open [7]. In revised versions

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of that paper, Knuth also recorded several subsequent developments, including a search-based even-case solution via ORTools CP-SAT [9], a GPT-5.3-codex construction for all even $m \geq 8$ [10], a Lean 4 formalization of the odd-case proof [8], and a particularly simple odd-case construction found through interaction with GPT 5.4 and Claude 4.6 Sonnet [4]. Ho Boon Suan later produced a complete proof of correctness for the codex-generated even-case construction [5]. More recently, a considerably simpler even-case construction was obtained by a multi-agent workflow combining symbolic reasoning, computational search, and cross-agent transfer of intermediate representations, although a symbolic proof of its correctness for all m remained open [1].

In this note, we present our simple alternative construction for the even case. Accompanying files contain complete AI-generated proof drafts both for this construction and for the construction of [1], and we briefly summarize the broader development process behind these results. It is important to verify the correctness of the proofs of these two constructions, and so in addition to verifying them with a number of different models, we also human-verified the intuition as well as several important steps in the proof. Importantly, for the construction of Keston Aquino-Michaels we also checked a full lean proof using different models.

1.1 Our Contribution

Our contributions are as follows.

1. **Mathematical.** We present a simple alternative construction for the even case of the Hamiltonian decomposition problem. In accompanying files, we also provide complete AI-generated proof drafts both for this construction and for the simple even-case construction of [1].

2. **Methodological.** We include a brief summary of the broader development workflow behind these results, including the decomposition of the proof task into subproblems and the iterative generation and checking of proof drafts for both constructions..

2 A Simple Alternative Construction

In this section we describe our simple alternative construction for even m. Throughout, $m \geq 10$ is even, and we write

$$h = \frac{m}{2}.$$

The small cases $m = 4, 6, 8$ were handled separately by computer search; the uniform construction below is the one that emerged for all larger even values.

2.1 How the construction was found

Our starting point was inspired by recent AI-assisted discovery workflows in which a large reasoning model is coupled with tree search and automated feedback [3]. The tree search algorithm, adapts the code mutation and search strategies described in recent literature[2]. We adopted this perspective at the level of *search-program discovery*: rather than using search as the final combinatorial solver itself, we used a tree-search-guided workflow to design and improve a search program for the Hamiltonian decomposition problem.

The program worked in fiber coordinates

$$s = i + j + k \pmod{m},$$

and searched over a structured family in which one distinguished fiber, later denoted by $\mathcal{F}_h$ with $h = m/2$, was treated as a *special fiber*, while all other fibers were assigned constant routing types. The special fiber carried a sparse permutation table, and the remaining fibers were divided into three consecutive blocks of constant types, with the block sizes chosen by a threshold search.

Each candidate was filtered automatically by global constraints. The program first computed the induced return maps on $\mathbb{Z}_m^2$ and discarded the candidate unless each color produced a single cycle there; surviving candidates were then lifted back to the full 3-dimensional torus and checked directly to verify that the three color classes formed a Hamiltonian decomposition.

A key feature of the candidate search program was its use of heuristics to restrict attention to sparse patterns from the outset. In particular, on the special fiber it searched only over sparse permutation tables, with most entries fixed to the identity and only a small geometrically structured exceptional set allowed to vary. From the resulting candidates, we selected one search-based program and provided that program itself to Gemini Deep Think. We asked it to analyze the structure of the procedure and extract the regularities implicit in it, with the goal of conjecturing a uniform closed-form rule. The construction stated below is the outcome of that process.

2.2 The construction

Let

$$V = \mathbb{Z}_m^3.$$

At each vertex $(i, j, k) \in V$ there are three outgoing directed edges,

$$(i, j, k) \to (i + 1, j, k), \qquad (i, j, k) \to (i, j + 1, k), \qquad (i, j, k) \to (i, j, k + 1),$$

with all coordinates understood modulo m. For each color $c \in \{0, 1, 2\}$ we write

$$F_c(i, j, k) \in \{0, 1, 2\}$$

for the chosen outgoing direction, where

$$0 = e_1, \qquad 1 = e_2, \qquad 2 = e_3.$$

We define the fiber index

$$s(i, j, k) \equiv i + j + k \pmod{m},$$

and for each $s \in \mathbb{Z}_m$ let

$$\mathcal{F}_s = \{(i, j, k) \in \mathbb{Z}_m^3 : \ i + j + k \equiv s \pmod{m}\}.$$

The distinguished fiber is

$$\mathcal{F}_h, \qquad h = \frac{m}{2}.$$

Constant routing on non-special fibers. On each non-special fiber, the routing is constant. We use the following three constant routing types:

$$\mathsf{A} = (0, 1, 2), \qquad \mathsf{B} = (1, 2, 0), \qquad \mathsf{C} = (2, 0, 1).$$

Thus, if a fiber has type A, then color 0 moves in direction 0, color 1 in direction 1, and color 2 in direction 2 at every vertex of that fiber; similarly for B and C.

The non-special fibers are listed in cyclic order

$$0, 1, 2, \dots, m-1,$$

with the special fiber h omitted. We assign type A to the first t_a non-special fibers, type B to the next t_b fibers, and type C to the remaining t_c fibers, where

$$t_a + t_b + t_c = m - 1.$$

The construction uses the following block sizes:

$$(t_a, t_b, t_c) = \left(\frac{m}{2} - 2,\ 2,\ \frac{m}{2} - 1\right) \qquad \text{if } m \equiv 0 \pmod 4,$$

and

$$(t_a, t_b, t_c) = \left(\frac{m}{2} - 3,\ 4,\ \frac{m}{2} - 2\right) \qquad \text{if } m \equiv 2 \pmod 4.$$

Routing on the special fiber. We now define the routing on the special fiber $\mathcal{F}_h$. Since on $\mathcal{F}_h$ we have

$$k \equiv h - i - j \pmod m,$$

we may identify $\mathcal{F}_h$ with $\mathbb{Z}_m^2$ using coordinates (i, j).

At each point of the special fiber we assign a permutation of $(0, 1, 2)$. We encode the six elements of S_3 as follows:

$$0 = [0,1,2], \qquad 1 = [0,2,1], \qquad 2 = [1,0,2], \qquad 3 = [1,2,0], \qquad 4 = [2,0,1], \qquad 5 = [2,1,0].$$

Thus a table

$$P : \mathbb{Z}_m^2 \to \{0,1,2,3,4,5\}$$

specifies the routing on the special fiber: if $P(i,j) = r$, then at the vertex

$$\big(i, j, h - i - j\big)$$

the triple

$$\big(F_0, F_1, F_2\big)$$

is the permutation encoded by r.

Case I: $m \equiv 0 \pmod 4$

The default value is

$$P(i,j) = 0$$

except at the following positions.

Anti-diagonal staircase. For $2 \le i \le m-3$,

$$P(i, m-1-i) = 2.$$

Right-boundary staircase. For $2 \le i \le m-3$,

$$P(i, m-1) = 5.$$

Top corrections.

$$P(0,0) = 2, \qquad P(0,m-2) = 5,$$
$$P(1,m-2) = 4, \qquad P(1,m-1) = 3.$$

Bottom corrections.

$$P(m-2,0) = 1, \qquad P(m-2,1) = 3, \qquad P(m-2,m-2) = 2, \qquad P(m-2,m-1) = 1,$$
$$P(m-1,0) = 0, \qquad P(m-1,1) = 4, \qquad P(m-1,m-2) = 5, \qquad P(m-1,m-1) = 0,$$

and for $2 \le j \le m-3$,

$$P(m-1,j) = 1.$$

Case II: $m \equiv 2 \pmod 4$

Again the default value is

$$P(i,j) = 0$$

except at the following positions.

Top rows.

$$P(0,0) = 2, \qquad P(0,m-1) = 5,$$
$$P(1,0) = 5, \qquad P(1,m-2) = 2, \qquad P(1,m-1) = 1.$$

Left-boundary staircase. For $2 \le i \le m-3$,

$$P(i,0) = 5.$$

Anti-diagonal staircase. For $2 \le i \le m-3$,

$$P(i,m-1-i) = 2.$$

Bottom rows.

$$P(m-2,0) = 1, \qquad P(m-2,1) = 3, \qquad P(m-2,m-1) = 2,$$
$$P(m-1,0) = 0, \qquad P(m-1,1) = 4, \qquad P(m-1,m-1) = 5,$$

and for $2 \le j \le m-2$,

$$P(m-1,j) = 1.$$

Formal definition of the full field. We can now define the full routing field

$$F = (F_0, F_1, F_2).$$

1. If $(i, j, k) \notin \mathcal{F}_h$, let
$$s \equiv i + j + k \pmod{m}.$$
According to the position of s among the non-special fibers, assign to $\mathcal{F}_s$ one of the constant types $\mathsf{A}, \mathsf{B}, \mathsf{C}$ using the block sizes t_a, t_b, t_c above. The values of (F_0, F_1, F_2) at (i, j, k) are then given by that constant type.

2. If $(i, j, k) \in \mathcal{F}_h$, then
$$k \equiv h - i - j \pmod{m},$$
so the point is identified with the two-dimensional coordinate (i, j) on the special fiber. We then evaluate the permutation table $P(i, j)$ from Case I or Case II above, according to the congruence class of m modulo 4, and define
$$\big(F_0(i, j, k), F_1(i, j, k), F_2(i, j, k)\big)$$
to be the permutation encoded by $P(i, j)$.

This completes the definition of the construction.

Informal interpretation. Almost all fibers are constant, and the only nontrivial fiber is $\mathcal{F}_h$. On that distinguished fiber, the construction is a sparse perturbation pattern organized around an anti-diagonal staircase together with a small number of boundary and corner corrections. The block sizes (t_a, t_b, t_c) determine how the constant fibers are distributed around the torus.

2.3 Comparison with prior and alternative constructions

It is useful to compare the present construction with the earlier simple even-case construction from [1]. The construction in [1] is organized in a layer-based way: for even m, it consists of $m - 2$ bulk layers of a constant type together with two exceptional repair layers, usually described as a staircase layer and a terminal layer. Its main strength is that it gives a compact global description and was verified computationally for a large range of even values of m. At the same time, that construction is expressed in terms of exceptional layers rather than a single sparse exceptional fiber, and a symbolic proof of its correctness for all m was left open.

By contrast, the construction studied in this section is organized around one distinguished fiber $\mathcal{F}_h$. All non-special fibers are constant, while the entire nontrivial behavior is concentrated on a sparse permutation table on $\mathcal{F}_h$. From a structural point of view, this makes the exceptional set more localized: instead of two globally nontrivial layers, one has a single special fiber together with a sparse set of corrections inside it.

Complete AI-generated proof drafts for both constructions are provided in accompanying files. The next section turns to the proof-development workflow used after a construction had already been specified.

3 LLM-Generated Proof Development

We now describe the workflow used to obtain formal proofs for the constructions introduced in Section 2. Rather than asking a language model to produce a complete proof in a single pass, we used an iterative proof-search pipeline powered by Gemini 3.1 Pro Preview. Given a fixed construction, the workflow proceeds by decomposing the proof task into subproblems, generating draft arguments section by section, assembling them into a complete document, and then iteratively checking and revising the result.

Our methodology is positioned as an extension of the AI-assisted scientific research techniques introduced by Woodruff et al. [11]. While their work also used iterative refinement and problem decomposition, we extend their framework to support autonomous, long-horizon proof generation. Specifically, we scale their collaborative methodology by integrating a combination of network-based inference scaling, maintenance of a persistent master proof document, and autonomous verification.

This extended generate-verify-revise loop relies on these mechanisms to handle the extreme length and structural constraints of the even-case construction:

Document Construction and Assembly: The system maintains a master proof document, decomposes the proof into subproblems, writes candidate arguments section by section, and assembles them into a single evolving draft.

Automatic Verification: Once assembled, an automatic verifier reviews the complete draft. Acting as a rigorous mathematical checker, it identifies unsupported claims, missing cases, or circular dependencies, and feeds this critique back into the next revision iteration.

Inference Scaling: To stabilize the generation of highly structured combinatorial arguments, network-based inference scaling is applied. The system leverages multiple candidate generations to produce stronger section drafts and more reliable verification feedback.

Through this extended framework, the system is able to search for a workable proof architecture, refine local arguments, and iteratively repair gaps until a formal proof emerges. This approach yielded the fully formal proof drafts presented in the accompanying files. More details will be released at a later date.

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